FOR ENGINEERS — THE LEDGER AN EPC CHECKS FIRST

THE ABSENCE AUDIT — THE METHOD, AT NATIONAL SCALE

Tunisia, audited — the engineer's ledger

the energy balance, water chemistry, acid-stream assays, lift floors, and every gate — everything a technical team verifies before FID.

Provenance law: MEASURED = cited anchor · DERIVED = arithmetic on measured values · ASSUMED = declared default. commercially-validated = 0 — no buyer has signed anything; published as zero on purpose. Generated 2026-09-09 by nations_audit.py / make_nations_pdfs.py — the same code that renders the page.

1. The energy ledger, leg by leg

Recovery ledger recomputed: 33.9 GWh/yr recovered + 20 GWh/yr dispatch value. The printed 82 GWh did not sum from its own legs (7.2 + 30 + 20 = 57.2; 24.8 unexplained) — corrected, with the arithmetic, on the page.

LEG	FORMULA	VALUE
Pump-as-turbines	9.81 × (70% × 150,000/86,400) m³/s × 50 m × 75% × 8,760 h	3.9 GWh/yr
Heat recovery	0.6 GJ/t × 2 Mt PG → 333 GWh thermal × 9%	30 GWh/yr
Flexibility	desal + storage follow the solar curve	20 GWh/yr (dispatch, not joules)
ERD	inside the 4 kWh/m³ SWRO benchmark (~35-40% of RO energy)	0 added — no double count

2. Water — the numbers an EPC checks first

PARAMETER	VALUE	ANCHOR
SWRO specific energy	4 kWh/m³ (benchmark incl. ERD)	Rw3 (mega-plant class)
Lift physics floor	1.09 kWh/m³ per 400 m at 100% eff; 1.45 at 75%	Rw4 — the physics that decides where water may travel
Delivered inland cost	$1.10-1.50/m³ vs irrigation tariffs $0.04-0.07	Rw6/Rw7 — inland water runs on brackish, not long-haul
SWRO CAPEX	$800-1,200 per m³/day (mega-plant)	Rw2
Gate	blended ≥$0.68/m³, ≥60% pre-sold, brine permit	computed (DSCR)

3. Materials — the acid stream

PARAMETER	VALUE	ANCHOR
REE content, Gafsa rock	322 mg/kg avg, peaks >1,700	Rm1

PARAMETER	VALUE	ANCHOR
PG partition of REE	60-85%	Rm2
Acid-stream U	85-90% of rock-U partitions to the wet-process acid	Ru1 (Freeport precedent)
U recovery (proven)	>95% DEHPA/TOPO, 14M lbs over 1978-99	Ru1
Radiology	Ra-226 214-970 Bq/kg in PG; exemption threshold 1,000 Bq/kg	Rm11/Rm12
Phase 0 assays	12 stream points: U, REE, Ra-226, Po-210, full metals	itemized in the audit

4. The dependencies, by status

DEPENDENCY	STATUS
Desal brine permit	**in design** — Gabes outfall / diffuser; permit required before SWRO FID
PG chemistry (Ra-226 partitioning)	**Phase 0 assay** — flowsheet keeps radium in gypsum (Freeport precedent); re-verify on Gafsa stream
Uranium handling	**gate** — CNSTN licensing, NORM transport, export controls
REE separation capex	**later phase** — +$244M solvent-extraction plant beyond the ~$100M circuit
Solar bankability	**gate** — DSCR ~0.9-1.0 at the record-low tariff; bankable region or grant support required
Grid connection	**later phase** — 3 GW connection + 400 kV upgrades ~$450M
Compute accelerators	**gate** — export-control licenses; colocation until an anchor tenant is licensed
Truffle yield	**gate** — 376 kg/ha is the 20-yr Murcia average; gate = 3-season record >=200 kg/ha
Export prices	**exposure** — U spot $80-95/lb; truffle EUR20-40/kg — no contracted offtake yet
Financing	**uncommitted** — DFI channel modeled for solar/desal; nothing committed

The water stack — line-by-line, with each credit's failure mode

The $0.347/m³ audit: the base is measured anchors; each credit is a loop-internal yield with a named failure mode; the degraded cascade still clears the $0.68 gate.

LINE	$/M³	PROVENANCE	WHAT MAKES IT FAIL
Base — solar energy 4 kWh/m³ at the loop's own PPA + capex + O&M	$0.567	MEASURED anchors (Rc13/Rw2/Rw3)	holds — if even the PPA lapses, grid power takes the base to $0.88, still under SONEDE
DC reject heat → RO feed preheat — one seawater pipe cools the servers, then arrives warm at the membranes (SEC −2.5%/°C)	−$0.007	DERIVED — Leg 03 + Leg 01, same pipework	fails only if no compute anchor tenant — and costs $0.007
Kiln/exotherm preheat — the acid plant and board kilns warm the same feed	−$0.000	DERIVED — Leg 04/05 waste heat	negligible by design — carried at zero margin

LINE	$\$/M^3$	PROVENANCE	WHAT MAKES IT FAIL
Brine valorization — salt/Mg evaporated on the loop's own free heat, sold at the park gate (of-take signed; nothing hauled)	−$0.137	DERIVED — brine is the desal's own by-product	if the of-take is not signed by FID, the credit degrades to −$0.03 (disposal avoided only) — the biggest single sensitivity
Shared civil works — one intake, one outfall, one grid connection serves RO + DC + metals in the same park	−$0.032	DERIVED — 12% capex sharing	if the park phases one module at a time, halve it to −$0.016
Dispatch value — the desal's own flexibility follows the loop's own solar curve	−$0.015	DERIVED — 20 GWh shifted	fails to zero if the STEG contract does not reward flexibility — contractual, not hoped
ESG financing delta — the health leg's public KPIs reprice the loop's own debt (100–150bp)	−$0.026	DERIVED — nations_v3	fails to zero if lenders do not accept the KPI track record — bankable-region fallback keeps the module alive
Brine pump-as-turbine — the desal's own residual pressure	−$0.002	DERIVED — 2.5 bar × 75%	fails to zero — immaterial
Integrated	**$\$0.347/\mathrm{m}^3$**	**28% of SONEDE's $1.00-1.50 (Pg16)**	**the degraded cascade: if every soft credit fails, water still lands at $\$0.521/\mathrm{m}^3$ — under SONEDE's own $1.00–1.50, and the $0.68 gate still clears with margin. The credits are the margin, not the plan.**

The people's energy is electrons, not hydrogen — and the H&sb2; lives inside the loop

The loop's H&sb2; (2.63 kt/yr) is an industrial molecule: per kWh of heat, green H&sb2; at $0.21 is 7× the loop's $0.031/kWh electron; blending caps at 5-20% by volume (Pg20/Pg21/Pg22); the whole H&sb2; volume equals 0.0% of grid-gas demand. The molecule goes where only a molecule works: ammonia for the GCT chain, inside the loop. the electrolyzer sits on the desal line, and everything it touches stays in the loop: the water it splits is the SWRO permeate (0.04% of the 150,000 m^3/d output); its un-split reject stream is clean water that returns to the product line; its oxygen by-product aerates the spirulina ponds and greenhouses; its stack heat (15–20% of input) joins the kiln and DC reject heat on the desal preheat circuit; and the minerals concentrated in its flush stream join the brine valorization line. The electrolyzer is not a consumer of the loop — it is another recovery point inside it.

The total routing — every stream gets a home

The ponds do *not* drink brine: SWRO brine is 55-70 g/L, past Arthrospira's ~30 g/L ceiling (Pg31); they drink the desal permeate and breathe the electrolyzer's oxygen. The full inventory:

STREAM	PRODUCED BY	ITS HOME	THE NUMBER
Solar electrons	500 MW fleet	desal, metals, DCs, electrolyzer, pumps — routed by the dispatcher	876 GWh/yr

STREAM	PRODUCED BY	ITS HOME	THE NUMBER
SWRO permeate	coastal desal	city offtake · electrolyzer feed · ponds · greenhouses	150,000 m^3/d
SWRO brine	coastal desal	salt + magnesium valorization, evaporated on the loop's own free heat	122,727 m^3/d → 58 kt Mg/yr
Brine bittern	valorization	Mg(OH)&sb2; fire retardant → **the board's own chemistry**	140 kt/yr = 140% of the board's need
Electrolyzer oxygen	H&sb2; plant	spirulina pond + greenhouse aeration	21 kt/yr
Electrolyzer hydrogen	H&sb2; plant	ammonia → the GCT fertilizer chain	2.63 kt/yr
Electrolyzer reject water / heat / flush	H&sb2; plant	product line / desal preheat / brine line	nothing leaves
DC reject heat	compute park	the same seawater pipe, arriving warm at the membranes	−\$0.007/m^3
Kiln + acid exotherm	board kilns + metals plant	desal preheat + greenhouses	30 GWh/yr
CO&sb2; (kiln + power)	combustion streams	mineralized into board (2 Mt PG) · pond carbon	540 t CO&sb2;/yr to the ponds
Phosphogypsum	GCT stack	fire-rated board — with its retardant from our own brine	2 Mt/yr
Acid stream	GCT wet-process acid	U + heavy REE extraction; raffinate back to fertilizer	716.5k lbs U&sb3;O&sb8;/yr
Municipal return flows	city mesh	30%+ reuse · pump-as-turbines on the descent	3.9 GWh/yr
Sludge / digestate	wastewater + organics	biogas for kilns and greenhouses	booked option, honestly labeled
Slaughterhouse bones	local slaughterhouses	bone char → the fluoride filters; **spent char → phosphate soil amendment for the fields**	filter loop closes into fertilizer
Spent pond medium	spirulina harvest	nutrient-rich → greenhouse fertigation	nothing discarded
Ammonia fertilizer	GCT chain	the terfas fields' host plants + greenhouse soils	the truffle grows on the loop's own nitrogen
Compute services	colo park	export — the data product	tenant-gated
Ra-226 / Po-210	acid stream radiology	stays in the gypsum/board matrix — the radiology gate	below the 1,000 Bq/kg exemption
Board offcuts	board line	recycled into new board	zero waste
Sodium chloride	brine line	industrial offtake at the park gate	nothing hauled
Halophytes / Artemia on brine	brine line	future option, honestly labeled — not a current credit	the only stream awaiting its consumer

the ponds drink the desal permeate — ~{:.0f} m³/d, {:.2f}% of the 150,000 m³/d output — and breathe the electrolyzer's oxygen; the brine (55–70 g/L, far past the ~30 g/L ceiling the organism tolerates) is routed to salt and magnesium valorization instead. Halophytes and Artemia on brine are booked as a future option, honestly labeled — not a current credit.

The fuel invoice, honestly split — the 500 MW is the grid's medicine, not the whole prescription

of the $4.895B fuel invoice, the 500 MW fleet displaces only the power-sector gas slice — ~$0.75B/yr (DERIVED from the energy balance) — by $106-125M/yr, which is 14133-16667% of that slice. The rest of the invoice has its own answers, not solar's: the bottles (~$0.55B of LPG, the bonbonne) leave the market as the people shift to electrons — induction on the loop's own surplus — and the biogas pilot; transport fuels (~$2.6B) are the honest frontier. The 500 MW is the grid's medicine, not the country's entire prescription.

The mountains, honestly analyzed

Fog collection is real where fog lives (10-20 L/m²/day, Pg33) — Tunisia's fog lives in the already-humid NW; the truffle mountains are the arid interior, and cloud seeding has no proven routine gain (Pg32). What the mountains give the loop: the mountains enter the loop as storage, rainfall and free cooling — and as the land where the truffle belongs (freeing the lowlands that grow today's food). Fog harvesting from pipe meshes stays off the program: the moisture is not there to collect, and the honest scientist does not plant a net under a cloudless sky.

The AI regulation — five layers, down to the drippers

The dispatcher is a model-predictive controller on a 15-minute receding horizon over the mass balances (Lexicographic: safety > contracts > economics); the twin trains it offline and audits it online, with a Kalman filter for unmeasured states and a published prediction-error KPI; node controllers stay PID-class with hard-wired interlocks; the governance layer never delegates a gate.

LAYER	WHAT IT DOES	THE MATHEMATICIAN'S NOTE
Layer 0 — the instrumented field	every node is a sensor + an actuator: membrane pressure and conductivity, brine concentration, pond pH and temperature, electrolyzer stack voltage, kiln thermocouples, PAT speed, reservoir levels, household-filter telemetry, the grid tie, the weather station	the stream inventory above is the measured state — the mass balances are the constraints the controller must never violate
Layer 1 — node controllers (edge, millisecond-to-second)	local closed loops of PID class: membrane pressure, pond dosing, kiln temperature, pump speed — with hard-wired safety interlocks	functional safety is never on the AI: a controller that fails safe is a relay, not a model

LAYER	WHAT IT DOES	THE MATHEMATICIAN'S NOTE
Layer 2 — the dispatcher (model predictive control, 15-minute receding horizon)	a multi-variable MPC: state = solar forecast, reservoir inventory, brine concentration, electrolyzer turndown, DC load, grid price, offtake commitments; decisions = pump flows, setpoints, brine routing, dispatch; forecasts 36–48 h weather, 24 h load	objective: minimize loop energy cost + degradation + gate-violation penalties, subject to mass balances, storage states and contract floors — lexicographic, never weighted: safety > contracts > economics. Its measured output is the 20 GWh/yr dispatch value and the routing-dependent water credits
Layer 3 — the digital twin (simulation, learning, audit)	a physics-calibrated model of every balance: trains the dispatcher offline, audits every decision after the fact, simulates before any physical change	unmeasured states (membrane fouling, pond bloom state) are estimated with a Kalman-class filter from pressure-flow and optical residuals; the twin's prediction error is a published KPI — the AI's accuracy is measured, not claimed
Layer 4 — the governance layer (the boundary, unchanged)	the scoreboard computes and publishes every gate number from Layer 0 telemetry with provenance; the price index; the results framework; anomaly flags to humans	the AI never sets a gate, never signs, never prices — the falsification ladder is mechanical thresholds, not a learned model. The nervous system is a servant, not a master

NODE	SENSES	CONTROLS	TIMESCALE	LAYER
PV field	irradiance, string current, panel temperature, soiling sensors	inverter setpoints, curtailment, cleaning scheduling	seconds–minutes	L1–L2
Desal trains	feed pressure, permeate conductivity, ERD pressure, CIP condition	high-pressure pump, ERD bypass, antiscalant dosing, CIP cycles	seconds–hours	L1–L2
Membrane health	pressure-flow residuals (fouling inferred by Kalman filter)	maintenance scheduling, train rotation	days	L3
Brine line	concentration, temperature, Mg saturation	evaporator feed rate, Mg(OH)&sb2; precipitation setpoints	minutes	L1–L2
Electrolyzer	stack voltage, current, purity, water feed	turndown, ramp rate, O&sb2; vent routing to ponds, reject routing	seconds	L1–L2
Board kilns	temperature profile, feed rate, offcut fraction	burner setpoints, retardant dosing, offcut recycle	minutes	L1–L2
Metals plant	SX stream assays, raffinate quality	extraction/stripping setpoints, raffinate return to fertilizer	minutes–hours	L1–L2
DC park	cooling-loop temperature, PUE, workload mix	seawater flow to racks, batch-workload scheduling to solar peaks	minutes	L2
Water mesh	reservoir levels, pump status, pressure heads, PAT speed	pump scheduling, zone valves, PAT bypass, the solar-following curve	15-minute dispatch	L2
Spirulina ponds	pH, salinity, temperature, optical density	aeration timing, CO&sb2; injection, medium exchange, harvest scheduling	hours	L1–L2

NODE	SENSES	CONTROLS	TIMESCALE	LAYER
Greenhouses	EC, pH, humidity, CO&sb2;, soil moisture	fertigation dosing (spent medium), ventilation, CO&sb2; enrichment from the kiln stream	minutes	L1–L2
Establishment drip — to the emitter	per-zone capacitance soil-moisture probes, per-zone flow meters, weather forecast	zone valve actuators per management block, pump pressure, fertigation dosing — emitters stay passive pressure-compensated; clogged drippers detected from per-zone flow residuals	hourly–daily	L1–L2
Household filters	usage telemetry, replacement flags	fleet logistics: replacement routing, spare inventory, the pilot's uptime KPI	days	L2–L4
Grid tie	import/export price, solar forecast, offtake commitments	import vs. dispatch decisions	15-minute	L2
Scoreboard + cohort	every KPI stream above, provenance-hashed	gate computation, price index, anomaly flags to humans	continuous	L4

what the AI is honestly worth: the 20 GWh/yr dispatch (~$0.8M/yr) + the routing-dependent water credits it executes (~$0.05-0.08/m^3 — DC heat, kiln heat, brine heat, dispatch) + a DERIVED 1–3% park OPEX from multi-variable coupling no manual schedule can follow — and the scoreboard, which is the program's entire credibility. if the AI is down, the loop does not stop: Layer 1 runs every node in safe, suboptimal mode — the degraded cascade already computed ($0.521/m^3 water, gate still clear). AI uptime is a KPI, not a dependency.

The irrigation — to the dripper, honestly

the fields drink at establishment, then live on rain — and the greenhouse loop runs on the pond's own spent nutrients. The irrigation system is designed the way the truffle itself is: drought-proof by architecture, not by subsidy. Establishment: 624 m^3/ha/yr for the first two years → 4,274 m^3/d for 2,500 ha, from inland brackish water and reuse; then rainfed.

Cloud generation — the honest version

our 12 km^2 array absorbs ~156 MW more sunlight than the bare soil (albedo Δ0.05) — a persistent local heat source, plus the ponds' evaporation placed upwind of the fields. Li et al. (Pg34) measured the class of effect at continental scale; at ours it is a designed microclimate, not a rain machine: soil shaded by panels retains moisture, the updraft adds marginal convective activity, and the dew that follows is free water on the host plants. The honest claim is directional and modest — and it costs nothing extra: the array was being built anyway.

The mountain portfolio — if mountain agriculture is needed, the version that scales

CROP / SYSTEM	HECTARES	WATER	VALUE	THE SCALE LOGIC
Terfas belt (arid jebels)	2,500-3,000 ha	rainfed + micro-catchments + dew	€15-45M/yr at scale (market-capped)	already the design: the drought-proof luxury crop

CROP / SYSTEM	HECTARES	WATER	VALUE	THE SCALE LOGIC
				on land nothing else farms
Cactus pear (opuntia) belt	consolidate + expand the existing ~600k ha national belt	zero irrigation; 300-500 mm survives on 150	fruit at 20-40 t/ha for the basin's price index + cladodes as livestock fodder	the arid mountain staple — Tunisia is already a world producer; the program adds value chains, not water
Medicinal & aromatic plants (rosemary, thyme, oregano)	5,000-10,000 ha across the Dorsale	rainfed; 300-500 kg/ha dry	~€8-20M/yr export at €4/kg dry	the highest value-per-kilo crop that needs no water at all — the mountain cash crop #2
Agrivoltaic fodder (sulla/alfalfa under the PV field)	up to 1,250 ha of shaded soil under the 500 MW array	the panels' shade halves evaporation; no added water beyond the field's own rain	fodder for ~5,000-10,000 head; the flock fertilizes the field	the loop's livestock leg: sheep graze solar parks already (Spain/US precedent) — the AI routes the grazing schedule
Rainwater harvesting (swales + terracing)	2-3% of mountain catchment area	each ha of swales captures ~3,000 m^3/yr at 300 mm rainfall	$200-500/ha — an order of magnitude under drip-system capex	the mountains' real water system: catch the rain that already falls there, at scale, without pipes

mountain agriculture at scale means putting each crop where its water cost is zero: terfas, cactus, MAPs and agrivoltaic fodder on the mountains' own rain and the loop's own shade — while wheat, vegetables and orchards stay on the lowlands and in the greenhouses where water is abundant. The mountains feed the people what only mountains can grow; the pipes carry water only where gravity and economics both say yes.

The growroom at landscape scale — climate steering toward bio-recovery

Measures: Atmospheric transect: RH / temperature / dew-point stations up the slope at the mist corridor, leaf-wetness sensors, 2-3 sonic anemometers for the anabatic flow; Soil grid: capacitance moisture probes per management zone (the drip grid doubled as the climate grid), soil temperature; The sky: satellite NDVI every 5 days into the twin, rain gauges along the corridor, pond evaporation pans; The park: panel albedo/soiling sensors (the field's reflectivity IS a climate actuator), pond water temperature

Steers: The mist corridor: fog nozzles, anabatic hours only, virtual-temperature-positive dosing — 2% of desal flow; The shade: the PV array's own footprint: shaded soil retains moisture; cleaning schedule shifts albedo; The pond surface: evaporation timing follows the wind — the ponds breathe upwind of the fields on purpose; The swales: rainfall harvested at contour, released by zone valves into the host-plant belt

Objective: the bio-recovery term in the dispatcher's objective: maximize dew days + soil-moisture persistence above wilting point + NDVI slope — subject to the energy budget, the thermals remaining buoyant, and the rain-gauge referee. The twin adds a soil-water-balance module (Penman-Monteith evapotranspiration, standard equations, computed in code) so every steering decision has a predicted and a measured outcome.

The spiral: the feedback the steering rides: mist → dawn dew → soil moisture → host plants establish → transpiration raises local humidity → more dew — the humidity loop that turns a slope back green, one feedback cycle at a time. The KPIs published: dew days per month, days above wilting point, NDVI slope — bio-recovery, measured like a growroom, audited like everything else.

The invisible drop — and the effects ledger, direct and side

the drop → microbial pulse → nutrient mineralization → plant-available N and P → host-plant growth → nectar and pollen → the insects: wild bees at 1,000-2,000 flower visits a day, ants engineering the soil, beetles on the wind → the host plants set seed (Helianthemum is insect-pollinated) → the mycorrhizal trade persists → the truffle → the harvest → the price index. Every dew morning the mist buys is 10&sup8; cells per square centimeter of leaf waking up — and the corridor's vegetation is the pollinator highway the truffle's own host plants depend on. One invisible drop feeds the microbes that feed the plants that feed the insects that pollinate the plants that feed the truffle that feeds the people.

and the drop re-enlists the army Tunisia lost: the right-of-way vegetation is the refuge for the beneficials — ladybirds (5,000 aphids each in a lifetime), lacewings, hoverfly larvae (400-600 aphids each), and the parasitoid wasps Tunisia already proved at national scale in the date oases (Trichogramma against the date moth). Every predator the corridor hosts is a pesticide that was not bought, not sprayed, and not inhaled by a child — biocontrol is the health ledger's quietest leg.

EFFECT	CLASS	THE MODEL	THE INSTRUMENT
Mist vapor transport	DIRECT	Gaussian plume + anabatic advection	sonic anemometers, RH transect
Dew deposition on host plants	DIRECT	surface energy-balance dew model (T&sb2;surf < dew point)	leaf-wetness sensors, dew gauges
Soil moisture gain, corridor zone	DIRECT	Penman-Monteith water balance	capacitance probe grid
Seedling establishment survival	DIRECT	water-stress survival curves	nursery registry, field counts
PV-shade moisture retention	DIRECT	shade fraction × PET reduction	soil probes under panels
Thermal buoyancy (the constraint)	DIRECT	virtual-temperature profile	T/RH + wind stations on the slope
Microbial Birch pulse	SIDE	rewetting respiration curves	soil CO&sb2; efflux chambers
Beneficial insect recruitment	SIDE	habitat-area × predator-prey dynamics	sticky traps, acoustic/AI counters
Pollinator visitation → host seed set	SIDE	pollination-saturation curves	visit counts, seed counts
Precipitation return	SIDE — ZERO-credited	cloud microphysics (the least-solved field, Pg32)	rain-gauge network
Pesticide displacement → child exposure	SIDE	exposure-response from the cohort	urinary pesticide metabolites

The land, projected — the Darwinian view of the environment itself

LAYER OF LIFE	WITHOUT THE PLAN	WITH THE PLAN
Soil biology	the seed bank dies with the soil — desertification's positive feedback: less vegetation, less humidity, less dew, less vegetation	the Birch pulse wakes with every dew morning; the microbial city rebuilds; 10 km of hyphae per gram of soil come back to trade
The insects	beneficials stay locally extinct; the pesticide treadmill accelerates; pollinators thin, host plants set less seed	the right-of-way is the refuge: ladybirds, lacewings, hoverflies, Trichogramma's cousins re-recruit; every predator is a pesticide not bought
The dew economy	bare, crusted ground sheds what little rain falls; the drop dies on arrival	the mist corridor turns the dawn into a harvest: dew on host plants, soil moisture banked, the spiral compounding
The truffle belt	the wild harvest shrinks; the mycorrhizal network starves	the belt establishes on the corridor's moisture: 100 ha, then 2,500-3,000 ha, on the land the humidity itself prepared
The frontier	desertification marches downhill — the slope's fitness landscape selects for stone	each humidified season pushes the establishment frontier a little further upslope — the landscape's selection flips from bare ground to vegetation

the Darwinian projection of the whole machine: the same land, two futures — one where the environment keeps selecting for stone, and one where a 2% mist margin re-engineers the fitness landscape so that life out-competes bare ground. The honest bound stands: the corridor transforms a belt, not a mountain range — the wider slopes change only through the slow, compounding spiral and the rain cycle. We plant the drop; the land grows the rest.

What this document is: a concept-level engineering design whose numbers are computed in code from sourced primitives — recomputed, corrected, and applied into the design. Free and public.

What it is not: a commissioned state program, a bank-grade feasibility study, an investment offer, or a claim that any of this has been built. Nothing here is advice to any government or any investor. Nothing replaces geological, radiological, geotechnical or EPC-level surveys. Verify every figure and citation before relying on any of it.

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11. [Rw3] LCOW at plant gate: $0.40-0.60/m³ (Jubail 3A $0.41; Sorek II bid $0.32-0.45); energy share 40-60% of OPEX — https://pmc.ncbi.nlm.nih.gov/articles/PMC8953854/...
12. [Rw4] Pumping physics: 400 m lift = 1.09 kWh/m³ at 100% eff; 1.45 at 75%; friction 0.5-1.5 kWh/m³ over 300 km; realistic 2.0-2.5 kWh/m³ — https://www.engineeringtoolbox.com/water-pumping-costs-d_1527.html...
13. [Rw5] Jubail-Buraydah IWTP: 587 km, 650,000 m³/d, $2.26B ($3.85M/km); twin-pipe scale $5-7M/km — https://www.meed.com/saudi-arabia-awards-22bn-water-transmission-contract...
14. [Rw6] Delivered inland desalinated cost $1.10-1.50/m³ (plant gate + transport + pumping energy) — https://investriyadh.ai/intelligence/water-scarcity-investment/...
15. [Rw7] Tunisian water tariffs: SONEDE 0.200-2.310 TND/m³; irrigation 0.060-0.160 TND/m³; groundwater 0.112-0.162 TND/m³ — https://www.sonede.com.tn/tout-savoir-sur-leau/la-facture-deau-methodes-de-paiement/...
16. [Rw13] Grid constraints: 3 GW overwhelms local 400 kV; new lines $1-2.5M/km; renewables <5% of generation; curtailment risk — https://reglobal.org/tunisia-focuses-on-grid-expansion-for-integrating-renewable-energy/...
17. [Rc2] Gafsa basin: DInSAR-documented subsidence at Moulares; geothermal gradient 25-30°C/km → 23-29°C at 100-300 m depth — https://www.researchgate.net/publication/251701016_Phosphate_mine_subsidences_deduced_from_differential_interf...
18. [Rc3] Gafsa radiation: 226Ra 16.3-375.1 Bq/kg in soils; phosphate tailings to 7,606 Bq/kg; radon-222 accumulation in enclosed workings; alpha-induced soft errors — https://www.researchgate.net/publication/380540277_Assessment_of_radiation_exposure_in_the_phosphate_mining_ar...
19. [Rc4] ASHRAE TC9.9 Class H1 (5th ed.): 18-22°C recommended, 25°C max for AI accelerators; AI datacenter CAPEX $20-35M/MW (incl. IT), $8-15M/MW facility-only — https://envigilance.com/compliance/ashrae-tc-9-9/...
20. [Rc5] GPU cost: ~100,000 B200-class per 100 MW; $25-40k each → $2.5-4B per 100 MW; 300 MW ≈ $7.5-12B — https://intuitionlabs.ai/articles/data-center-gpu-pricing-2026...
21. [Rc6] US BIS export controls 2024-26: UAE 35,000 GPUs (G42/HUMAIN, 'Compute Diplomacy'); Saudi case-by-case 35,000 cap; Tunisia has no bilateral AI-security agreement — https://www.commerce.gov/news/press-releases/2025/11/statement-uae-and-saudi-chip-exports...
22. [Ru1] Uranium from wet-process phosphoric acid: Freeport (US) produced 14M lbs U3O8 1978-99 at >95% DEHPA/TOPO recovery; ~20% of US uranium in the 1980s came from phosphates — https://www.osti.gov/servlets/purl/1357599...
23. [Ru3] WPA uranium SX cost $59-83/lb U3O8 (IX $46-76) vs $80-95/lb 2024-26 spot — https://repositories.lib.utexas.edu/bitstreams/89897275-85cf-4cd6-a0c5-1d8daa353799/download...
24. [Ru4] REE from phosphoric acid: PhosAgro pilot; D2EHPA extracts >85% of heavy REEs (Dy, Y, Er) from WPA — https://www.mdpi.com/2673-4117/7/2/58...
25. [Ru5] Gafsa phosphate rock uranium content 45-140 ppm (southern basin) — https://www.researchgate.net/publication/380540277_Assessment_of_radiation_exposure_in_the_phosphate_mining_ar...